

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286295

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

KOBAYASHI; Atsushi et al.

CONNECTOR

Abstract

A connector comprises a hold-down and a housing which has a protected portion. The hold-down has a base portion, two side arms which are connected to opposite sides of the base portion in a width direction, respectively, and a middle arm which extends inward in a pitch direction from the base portion. Each of the side arms has a side portion which extends along the pitch direction, an extending portion which extends inward in the width direction from a lower end of the side portion and a spring portion which extends upward from the extending portion so as to be resiliently deformable. The middle arm has a protector. The protector covers the protected portion. The protected portion is formed with movement-allowance portions which allow movements of the spring portions in accordance with resilient deformations thereof, respectively. The protector is located above the movement-allowance portions.

Inventors: KOBAYASHI; Atsushi (Tokyo, JP), OMODACHI; Takashi (Tokyo, JP)

Applicant: JAPAN AVIATION ELECTRONICS INDUSTRY, LIMITED (Tokyo, JP)

Family ID: 1000008475361

Assignee: JAPAN AVIATION ELECTRONICS INDUSTRY, LIMITED (Tokyo, JP)

Appl. No.: 19/036207

Filed: January 24, 2025

Foreign Application Priority Data

JP	2024-034359	Mar. 06, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: H01R12/91 (20110101); H01R12/70 (20110101); H01R13/502 (20060101);
H01R13/629 (20060101)

U.S. Cl.:

CPC **H01R12/91** (20130101); **H01R12/707** (20130101); **H01R13/5025** (20130101);
H01R13/629 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. JP 2024-034359 filed Mar. 6, 2024, the content of which is incorporated herein in its entirety by reference.

BACKGROUND OF THE INVENTION

[0002] This invention relates to a connector comprising a hold-down.

[0003] For example, this type of connector is disclosed in WO2018/236070A (Patent Document 1), the content of which is incorporated herein by reference.

[0004] Referring to FIG. 15, Patent Document 1 discloses a connector **90** comprising an insulator body (housing) **92** and two guide members (hold-downs) **94**. The hold-downs **94** are attached to opposite ends of the housing **92** in a pitch direction, respectively. The hold-downs **94** reinforce the housing **92** and guide a mating connector **98** when the mating connector **98** is mated with the connector **90**. Referring to FIG. 15 together with FIG. 16, each of the hold-downs **94** has two contact portions (spring portions) **96**. Each of the spring portions **96** works as a terminal through which a large current flows under a mated state where the connector **90** and the mating connector **98** are mated with each other.

[0005] The spring portion **96** might be brought into abutment with the mating connector **98** to buckle during a process in which the mating connector **98** is mated with the connector **90**.

SUMMARY OF THE INVENTION

[0006] It is therefore an object of the present invention to provide a connector with a structure which can prevent a spring portion of a hold-down from buckling.

[0007] An aspect of the present invention provides a connector comprising a housing and a hold-down. The housing has two long walls, two short walls, an island-like portion and a bottom portion. Each of the long walls extends along a pitch direction. Each of the short walls couples the long walls together in a width direction perpendicular to the pitch direction. The island-like portion is enclosed by the long walls and the short walls in a horizontal plane defined by the pitch direction and the width direction. The bottom portion is located at a lower side of the housing in an up-down direction perpendicular to both the pitch direction and the width direction. The bottom portion couples the island-like portion to each of the long walls and the short walls. The island-like portion has a protected portion. The protected portion is located at an end of the island-like portion in the pitch direction. The hold-down has a base portion, two side arms and a middle arm. The base portion has a cover portion. The cover portion covers the short wall from above. The two side arms are connected to opposite sides of the base portion in the width direction, respectively. Each of the side arms has a side cover, a side portion, an extending portion, a spring portion and a contact point. For each of the side arms, the side cover covers the long wall from above, the side portion is connected to an outer part of the side cover in the width direction and extends along the pitch direction, the extending portion extends inward in the width direction from a lower end of the side portion, the spring portion extends upward from the extending portion so as to be resiliently deformable, and the contact point is supported by the spring portion. The middle arm has a middle portion and a protector. The middle portion extends inward in the pitch direction from the base portion. The protector extends from the middle portion and covers the protected portion of the

island-like portion. The protected portion is formed with two movement-allowance portions. The two movement-allowance portions are recessed inward in the width direction and allow movements of the spring portions in accordance with resilient deformations thereof, respectively. The protector is located above the two movement-allowance portions. Each of the side arms is formed with a bent portion which is located at a boundary between a lower end of the side portion and the extending portion.

[0008] The extending portion of the hold-down according to an aspect of the present invention can be formed by bending a metal plate, which partially forms the side portion, inward in the width direction at a lower end of the side portion. The boundary between the extending portion and the side portion is the bent portion which is formed as described above and can be securely soldered on a board. Moreover, since the spring portion according to an aspect of the present invention can be arranged in the vicinity of the island-like portion, the long wall does not need to be formed with a space which allows a resilient deformation of the spring portion. Accordingly, the housing can be improved in its strength.

[0009] The protector of an aspect of the present invention is arranged above the spring portions so that the spring portions are protected from above. The thus-arranged protector prevents the spring portions from being brought into abutment with a mating connector during a process in which the mating connector is mated with the connector, for example. Thus, an aspect of the present invention provides a connector with a structure which can prevent a spring portion of a hold-down from buckling.

[0010] An appreciation of the objectives of the present invention and a more complete understanding of its configuration may be had by studying the following description of the preferred embodiment and by referring to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a perspective view showing a connector according to an embodiment of the present invention.

[0012] FIG. 2 is an enlarged, perspective view showing a part of the connector enclosed by dashed line A of FIG. 1.

[0013] FIG. 3 is a side view showing the connector of FIG. 1, wherein hidden outlines of a base portion and side arms of a hold-down are illustrated with dashed line.

[0014] FIG. 4 is a top view showing the connector of FIG. 1, wherein a part of the connector enclosed by dashed line is enlarged and illustrated.

[0015] FIG. 5 is a side view showing the connector of FIG. 1 together with a mating connector, wherein an outline of an upper surface of a board on which the connector is mounted and an outline of a lower surface of a mating board on which the mating connector is mounted are illustrated with dashed line.

[0016] FIG. 6 is a perspective view showing the mating connector of FIG. 5.

[0017] FIG. 7 is a cross-sectional view showing the connector of FIG. 4, taken along line VII-VII.

[0018] FIG. 8 is a cross-sectional view showing the connector of FIG. 4, taken along line VIII-VIII.

[0019] FIG. 9 is a cross-sectional view showing the connector of FIG. 4, taken along line IX-IX.

[0020] FIG. 10 is a perspective view showing the hold-down of FIG. 1, wherein an outline of a housing is partially illustrated with dashed line.

[0021] FIG. 11 is a cross-sectional view showing the connector of FIG. 4, taken along line XI-XI, wherein a middle part of the connector in the pitch direction is not illustrated.

[0022] FIG. 12 is another perspective view showing the hold-down of FIG. 10.

[0023] FIG. 13 is a top view showing the hold-down of FIG. 10, wherein a hidden outline of a flat-

plate portion of the base portion is illustrated with dashed line, and boundaries among the base portion, the side arms and a middle arm are illustrated with chain dotted line.

[0024] FIG. **14** is a side view showing the hold-down of FIG. **10**, wherein boundaries among the base portion and the side arms are illustrated with dashed line.

[0025] FIG. **15** is a perspective view showing a connector and a mating connector of Patent Document 1.

[0026] FIG. **16** is a perspective view showing a hold-down of the connector of FIG. **15**.

[0027] While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION

[0028] Referring to FIG. **1** together with FIG. **5**, a connector **10** according to an embodiment of the present invention is a so-called on-board connector which is configured to be mounted on a board **82**. The connector **10** is mateable with a mating connector **70**, which is mounted on a mating board **84**, along an up-down direction. The up-down direction of the present embodiment is the Z-direction. In the present embodiment, “upward” means the positive Z-direction, and “downward” means the negative Z-direction. The words such as the up-down direction do not indicate the absolute positional relation relative to the ground but merely indicate a relative positional relation in figures.

[0029] Referring to FIG. **1**, the connector **10** of the present embodiment comprises a housing **20** made of insulator, a plurality of terminals **50** each made of conductor and two hold-downs **60** each made of conductor. The hold-downs **60** are attached to opposite ends of the housing **20** in a pitch direction perpendicular to the up-down direction, respectively, and reinforce the housing **20**. The pitch direction of the present embodiment is the Y-direction. In the present embodiment, “inward in the pitch direction” means “toward the middle of the connector **10** in the pitch direction”, and “outward in the pitch direction” means “away from the middle of the connector **10** in the pitch direction”. As described above, the connector **10** has opposite ends to which the hold-downs **60** are attached. The opposite ends of the connector **10** have a mirror symmetric structure with respect to a vertical plane (XZ-plane) perpendicular to the pitch direction.

[0030] The connector **10** of the present embodiment has only the aforementioned members and has the aforementioned structure. However, the present invention is not limited thereto. For example, the connector **10** may further comprise another member in addition to the aforementioned members. The opposite ends of the connector **10** may have mirror unsymmetric structure with respect to the XZ-plane. When the opposite ends of the connector **10** are formed as described above, the number of the hold-downs **60** may be one.

[0031] Hereafter, explanation will be made about the mating connector **70** (see FIG. **6**) of the present embodiment.

[0032] Referring to FIG. **6** together with FIG. **5**, the mating connector **70** of the present embodiment comprises a mating housing **72** made of insulator, a plurality of mating terminals **76** each made of conductor and two mating hold-downs **78** each made of conductor. The mating hold-downs **78** are attached to opposite ends of the mating housing **72** in the pitch direction, respectively, and reinforce the mating housing **72**. The mating terminals **76** are provided so as to correspond to the terminals **50**, respectively.

[0033] Referring to FIG. **5**, under a mated state where the connector **10** and the mating connector **70** are mated with each other, each of the terminals **50** is brought into contact with the corresponding mating terminal **76**, and thereby the board **82** and the mating board **84** are electrically connected with each other. The hold-downs **60** are brought into contact with the mating

hold-downs **78**, respectively, under the mated state. The mating connector **70** of the present embodiment has the aforementioned structure. However, the structure of the mating connector **70** of the present invention is not specifically limited.

[0034] Hereafter, explanation will be made about the housing **20** of the present embodiment.

[0035] Referring to FIG. **1**, the housing **20** of the present embodiment is a unitary member mold of resin and has a peripheral wall **22**, an island-like portion **26** and a bottom portion **29**. The peripheral wall **22** has two long walls **23** and two short walls **24**. Thus, the housing **20** has the two long walls **23** and the two short wall **24**.

[0036] Each of the long walls **23** of the present embodiment extends along the pitch direction. Each of the short walls **24** of the present embodiment extends along a width direction perpendicular to both the up-down direction and the pitch direction and couples the two long walls **23** together in the width direction. The width direction of the present embodiment is the X-direction. In the present embodiment, “left-side” means the positive X-side, and “right-side” means the negative X-side. In the present embodiment, “inward in the width direction” means “toward the middle of the connector **10** in the width direction”, and “outward in the width direction” means “away from the middle of the connector **10** in the width direction”. Each of the long walls **23** has a size in the pitch direction which is larger than a size of the short walls **24** in the width direction. In other words, the long wall **23** is longer than the short wall **24**.

[0037] The peripheral wall **22** of the present embodiment is formed of the long walls **23** and the short walls **24** and has a seamless rectangular shape in a horizontal plane (XY-plane) defined by the pitch direction and the width direction. However, the present invention is not limited thereto, but the structure of the peripheral wall **22** can be modified as necessary.

[0038] The island-like portion **26** of the present embodiment is enclosed by the long walls **23** and the short walls **24** in the horizontal plane and extends along the pitch direction. The island-like portion **26** is formed with a middle hole **27**. The middle hole **27** is located at the middle of the island-like portion **26** in the width direction. The middle hole **27** extends along the pitch direction and passes through the island-like portion **26** in the up-down direction. As a result, the island-like portion **26** is formed with two parts which are located at opposite sides of the middle hole **27** in the width direction, respectively. The island-like portion **26** of the present embodiment has the aforementioned structure. However, the present invention is not limited thereto. For example, the middle hole **27** may be formed as necessary.

[0039] The bottom portion **29** of the present embodiment is located at a lower side of the housing **20**. The bottom portion **29** couples the island-like portion **26** to each of the long walls **23** and the short walls **24**. The island-like portion **26** of the present embodiment has two protected portions **28**. The protected portions **28** are located at opposite ends of the island-like portion **26** in the pitch direction, respectively. However, the present invention is not limited thereto. For example, the island-like portion **26** may have only one of the protected portions **28**. In this instance, the protected portion **28** may be located at an end of the island-like portion **26** in the pitch direction.

[0040] Referring to FIGS. **2**, **4** and **9**, each of the protected portions **28** of the present embodiment is formed with two movement-allowance portions **282**. The two movement-allowance portions **282** are located at opposite sides of the protected portion **28** in the width direction, respectively. Each of the movement-allowance portions **282** is a recess which is recessed inward in the width direction. In other words, the two movement-allowance portions **282** are recessed inward in the width direction.

[0041] Referring to FIG. **4** together with FIGS. **7** and **8**, the housing **20** is formed with a plurality of holding portions **32** which correspond to the terminals **50**, respectively. Each of the holding portions **32** is a space which vertically passes through the housing **20**. As described later, each of the holding portions **32** holds the corresponding terminal **50**.

[0042] Hereafter, explanation will be made about the terminals **50** of the present embodiment.

[0043] The terminals **50** of the present embodiment consist of a plurality of first terminals **50F** and

a plurality of second terminals **50S**. As can be seen from FIGS. **7** and **8**, the second terminal **50S** has a shape of the first terminal **50F** which is rotated by 180 degrees about an axis extending in parallel to the up-down direction. Thus, the first terminal **50F** and the second terminal **50S** are components same as each other although their arranged postures are different from each other. However, the present invention is not limited thereto. For example, the first terminal **50F** and the second terminal **50S** may have structures different from each other.

[0044] Each of the terminals **50** of the present embodiment is a so-called bent contact which is formed by bending a single metal plate. Thus, each of the terminals **50** is a single metal plate with bends. Each of the terminals **50** of the present embodiment has a fixed portion **52**, a held portion **54**, a spring portion **56** and two contact points **58**. Each of the terminals **50** has a structure as described below.

[0045] Referring to FIG. **5**, the fixed portion **52** is configured to be fixed on the board **82** via soldering. As shown in FIGS. **7** and **8**, the held portion **54** extends from the fixed portion **52**. The held portion **54** is provided with two press-fit projections (not shown). The press-fit projections project from opposite sides of the held portion **54** in the pitch direction, respectively. The spring portion **56** extends from the held portion **54**. The spring portion **56** has a U-like shape as a whole and is resiliently deformable easily. The two contact points **58** are located on opposite sides of the spring portion **56** in the width direction, respectively, and face each other in the width direction. Each of the two contact points **58** is supported by the spring portion **56** and is movable in accordance with a resilient deformation of the spring portion **56**.

[0046] Each of the terminals **50** of the present embodiment has the aforementioned structure. However, the present invention is not limited thereto, but the structure of each of the terminals **50** can be modified as necessary. For example, the number of the contact points **58** may be one.

[0047] Referring to FIGS. **7** and **8** together with FIG. **4**, the held portion **54** of each of the terminals **50** is press-fit into the corresponding holding portion **32** from below and is thereby securely held. Thus, each of the holding portions **32** holds the held portion **54** of the corresponding terminal **50**.

[0048] Referring to FIG. **4**, the terminals **50** of the present embodiment are divided into two terminal rows **50R**. The terminals **50** of each of the terminal rows **50R** are arranged along the pitch direction. Each of the terminal rows **50R** includes a plurality of the first terminals **50F** and a plurality of the second terminals **50S**. The first terminals **50F** and the second terminals **50S** of each of the terminal rows **50R** are alternately arranged at regular intervals in the pitch direction.

[0049] The two terminal rows **50R** are arranged along a left side and a right side of the housing **20** in the width direction, respectively. Each of the first terminals **50F** of the left-side terminal row **50R** is held by the holding portion **32** formed in the left-side long wall **23**. Each of the second terminals **50S** of the left-side terminal row **50R** is held by the holding portion **32** formed in the left-side part of the island-like portion **26**. Each of the first terminals **50F** of the right-side terminal row **50R** is held by the holding portion **32** formed in the right-side part of the island-like portion **26**. Each of the second terminals **50S** of the right-side terminal row **50R** is held by the holding portion **32** formed in the right-side long wall **23**.

[0050] Referring to FIGS. **7** and **8**, the first terminal **50F** and the second terminal **50S** which are adjacent to each other in the pitch direction are arranged so that the fixed portions **52** thereof face away from each other in the width direction. As a result of this arrangement, the fixed portions **52** of any two of the terminals **50** adjacent to each other in the pitch direction, which are the first terminal **50F** and the second terminal **50S**, are located at positions different from each other in the width direction. Thus, the terminals **50** of the present embodiment are arranged in an alternate arrangement (staggered arrangement) so that the terminals **50** can be arranged in a narrow pitch.

[0051] The terminals **50** of the present embodiment are arranged as described above. However, the present invention is not limited thereto. For example, each of the terminal rows **50R** may include only the first terminals **50F** or include only the second terminals **50S**. In other words, the terminals **50** do not need to be arranged in a staggered arrangement.

[0052] Referring to FIG. 4, according to the present embodiment, it is possible to visually inspect whether the fixed portions **52** are properly soldered on the board **82** (see FIG. 5) or not through the middle hole **27** since the island-like portion **26** is formed with the middle hole **27**. However, the present invention is not limited thereto. The middle hole **27** does not need to be formed in an instance in which the terminals **50** are not arranged in a staggered arrangement, for example.

[0053] Hereafter, explanation will be made about the hold-downs **60** (see FIG. 1) of the present embodiment.

[0054] Referring to FIG. 1, the two hold-downs **60** of the present embodiment are provided so as to correspond to the two short walls **24** of the housing **20**, respectively. Each of the hold-downs **60** is insert-molded in the housing **20** and is thereby securely fixed to the housing **20**. However, the fixing method of each of the hold-downs **60** to the housing **20** is not specifically limited.

[0055] One of the two hold-downs **60** of the present embodiment has a shape of a remaining one of the two hold-downs **60** which is rotated by 180 degrees about an axis extending in parallel to the up-down direction. Thus, the two hold-downs **60** are components same as each other although their arranged postures are different from each other. Each of the hold-downs **60** has a mirror symmetric shape with respect to the YZ-plane. However, the present invention is not limited thereto. For example, the two hold-downs **60** may have structures different from each other.

[0056] Hereafter, explanation will be made about one of the hold-downs **60** of the present embodiment. The explanation described below is applicable to each of the hold-downs **60**.

[0057] Referring to FIGS. 10 and 12 together with FIG. 2, the hold-down **60** of the present embodiment is formed by bending a single metal plate. In other words, the hold-down **60** is a single metal plate with bends. The hold-down **60** has a base portion **62**, two side arms **66** and a middle arm **68**. The two side arms **66** are connected to opposite sides of the base portion **62** in the width direction, respectively. The middle arm **68** is connected to an inner part of the base portion **62** in the pitch direction. The hold-down **60** of the present embodiment is formed of the aforementioned portions and has the aforementioned structure. Each of the aforementioned portions is a part of the single hold-down **60**. However, the present invention is not limited thereto. For example, the hold-down **60** may further have another portion in addition to the aforementioned portions.

[0058] Referring to FIGS. 3 and 12, the base portion **62** of the present embodiment has a cover portion **63**, a flat-plate portion **64** and two fixed portions **65**.

[0059] Referring to FIGS. 2 and 11, the cover portion **63** covers the short wall **24** from above. In detail, the cover portion **63** is attached to the short wall **24** so as to straddle the short wall **24** in the pitch direction. The thus-attached cover portion **63** has an outer cover **632** which is located on an outer surface of the short wall **24** in the pitch direction and an inner cover **634** which is located on an inner surface of the short wall **24** in the pitch direction.

[0060] Referring to FIGS. 3 and 11, the flat-plate portion **64** is connected to a lower end of the outer cover **632**. The flat-plate portion **64** has a flat-plate shape perpendicular to the pitch direction and extends along the width direction. The most part of the flat-plate portion **64** is embedded in the short wall **24**.

[0061] Referring to FIGS. 3 and 14, the two fixed portions **65** are connected to opposite ends of the flat-plate portion **64** in the width direction, respectively. Each of the fixed portions **65** is exposed outward from the short wall **24** and extends downward from the flat-plate portion **64**. Referring to FIG. 12 together with FIG. 5, each of the fixed portions **65** has a lower end which is bent outward in the pitch direction and is configured to be fixed on the board **82** via soldering.

[0062] Referring to FIG. 11, the base portion **62** of the present embodiment has the aforementioned structure and reinforces the short wall **24**. However, the present invention is not limited thereto. For example, the flat-plate portion **64** and the fixed portions **65** may be provided as necessary.

[0063] Referring to FIG. 10, each of the side arms **66** of the present embodiment has a coupling portion **661**, a side cover **662**, a side portion **663**, an additional fixed portion **665**, an extending portion **667**, a spring portion **668** and a contact point **669**. Hereafter, explanation will be made

about one of the side arms **66** of the present embodiment. The explanation described below is applicable to each of the side arms **66**.

[0064] Referring to FIG. **10** together with FIG. **2**, the side cover **662** covers the long wall **23** from above. In detail, the side cover **662** is attached to the long wall **23** so as to straddle the long wall **23** in the width direction. The thus-attached side cover **662** has an outer part which is located on an outer surface of the long wall **23** in the width direction and an inner part which is located on an inner surface of the long wall **23** in the width direction. The inner part of the side cover **662** extends downward to a position located in the vicinity of the bottom portion **29** of the housing **20**.

[0065] The side portion **663** is connected to the outer part of the side cover **662** in the width direction. In detail, the side portion **663** is connected to a lower end of the outer part of the side cover **662** which is located at an outer side of the side cover **662** in the width direction. The side portion **663** has a flat-plate shape perpendicular to the width direction and extends along the pitch direction. The side portion **663** covers an outer surface of the long wall **23** in the width direction. The side portion **663** of the present embodiment is formed with a cut **664** which is located at the middle thereof in the pitch direction. The side portion **663** has two parts which are located at opposite sides of the cut **664** in the pitch direction, respectively.

[0066] The additional fixed portion **665** is connected to a lower end of a part of the side portion **663** which is located outward of the cut **664** in the pitch direction. The additional fixed portion **665** extends downward from the side portion **663**. Referring to FIG. **12** together with FIG. **5**, the additional fixed portion **665** has a lower end which extends straight along the up-down direction and is configured to be fixed on the board **82** via soldering.

[0067] Referring to FIG. **2**, each of the side cover **662**, the side portion **663** and the additional fixed portion **665** of the side arm **66** of the present embodiment has the aforementioned structure and reinforces the long wall **23**. However, the present invention is not limited thereto. For example, the additional fixed portion **665** may be provided as necessary.

[0068] Referring to FIGS. **10** and **12**, the coupling portion **661** of the present embodiment couples the base portion **62** and the side portion **663** together. In detail, referring to FIGS. **3** and **14** together with FIG. **10**, the coupling portion **661** extends outward in the width direction from an outer end of the base portion **62** in the width direction along the short wall **24** and then extends along the long wall **23** to an outer end of the side portion **663** in the pitch direction. Referring to FIG. **3** together with FIG. **14**, a part of the coupling portion **661** which extends along the short wall **24** is embedded in the short wall **24**.

[0069] Referring to FIG. **10**, the side arm **66** of the present embodiment is coupled to the base portion **62** by the coupling portion **661**. However, the present invention is not limited thereto. For example, the side arm **66** may be coupled to the base portion **62** by an alternative portion instead of the coupling portion **661**. This alternative portion may couple an upper end of the cover portion **63** and an upper end of the side cover **662** together. However, according to this structure, the cover portion **63** and the side cover **662** are inevitably formed of two predetermined parts of a single metal plate, respectively, the two predetermined parts being adjacent to each other in each of the width direction and the pitch direction. This forming method will restrict design flexibility. For example, in an instance in which the connector **10** has a small size in the width direction, a size of the inner part of the side cover **662**, which is located at an inner side of the side cover **662** in the width direction, cannot be made sufficiently long in the up-down direction.

[0070] In contrast, each of the cover portion **63** and the side cover **662** of the present embodiment can be formed in various shapes by bending a part of a single metal plate which extends along the width direction. Accordingly, the present embodiment is preferable from a viewpoint of improvement of design flexibility.

[0071] Referring to FIGS. **9** and **10**, the extending portion **667** extends inward in the width direction from a lower end of the side portion **663**. In detail, the extending portion **667** is connected to a lower end of a part of the side portion **663** which is located inward of the cut **664** in the pitch

direction. The extending portion **667** passes under the short wall **24** and then extends to a position located in the vicinity of the protected portion **28** of the island-like portion **26**.

[0072] The spring portion **668** is connected to an inner end of the extending portion **667** in the width direction. The spring portion **668** extends upward from the extending portion **667** so as to be resiliently deformable. According to the present embodiment, no part of the housing **20** is formed about a lower end the spring portion **668**. The thus-arranged lower end of the spring portion **668** is exposed upward and downward from the housing **20**. Accordingly, the spring portion **668** can be easily bent. However, the present invention is not limited thereto. For example, the lower end of the spring portion **668** may be embedded in the bottom portion **29** of the housing **20**.

[0073] The contact point **669** is supported by the spring portion **668** and is movable in the width direction in accordance with a resilient deformation of the spring portion **668**. The contact point **669** is easily movable inward in the width direction in accordance with a resilient deformation of the spring portion **668** since the protected portion **28** is formed with the movement-allowance portions **282**. Referring to FIG. 2, the contact point **669** is provided on an arc-like part of the spring portion **668** which protrudes outward in the width direction. The thus-provided contact point **669** faces outward in the width direction. The contact point **669** and the contact points **58** of some of the terminals **50** which face outward in the width direction are arranged in a straight line in the pitch direction. Referring to FIG. 9 together with FIG. 6, the contact point **669** of the present embodiment is brought into contact with the mating hold-down **78** of the mating connector **70** and works as a power contact under the mated state. However, the present invention is not limited thereto, but the usage of the spring portion **668** can be modified as necessary.

[0074] Referring to FIG. 10, each of the extending portion **667**, the spring portion **668** and the contact point **669** of the side arm **66** of the present embodiment has the aforementioned structure. If the spring portion **668** is arranged in the vicinity of the long wall **23**, the long wall **23** should be formed with a space which allows a resilient deformation of the spring portion **668**. The thus-formed space might degrade strength of the housing **20**. In contrast, according to the present embodiment, the long wall **23** does not need to be formed with a space which allows a resilient deformation of the spring portion **668** since the spring portion **668** can be arranged in the vicinity of the island-like portion **26**. Accordingly, the housing **20** can be improved in its strength.

[0075] According to the present embodiment, the side arm **66** is formed with a bent portion **666** which is located at a boundary between the lower end of the side portion **663** and the extending portion **667**. The extending portion **667** of the hold-down **60** can be formed by bending a metal plate, which partially forms the side portion **663**, inward in the width direction at a lower end of the side portion **663**. Referring to FIG. 9 together with FIG. 5, the boundary between the extending portion **667** and the side portion **663** is the bent portion **666** which is formed as described above. When the thus-formed boundary is soldered on the board **82**, a solder fillet is formed on the boundary, and thereby the bent portion **666** can be securely soldered on the board **82**.

[0076] Referring to FIG. 10, the cut **664** formed in the side portion **663** divides the side portion **663** into two parts which consist of a part formed with the side cover **662** and another part formed with the extending portion **667**. However, the present invention is not limited thereto. For example, the side cover **662** may be formed so as to extend over all the side portion **663** without forming the cut **664**, and the extending portion **667** may be connected to the whole of the lower end of the side portion **663**. However, according to this structure, it is difficult to bent the side cover **662** downward while the extending portion **667** is bent inward in the width direction. Accordingly, the present embodiment is preferable from a viewpoint of easy formation of the side cover **662** and the extending portion **667**.

[0077] Referring to FIGS. 2 and 11, the middle arm **68** of the present embodiment has a middle portion **682** and a protector **684**. The middle portion **682** is connected to a lower end of the inner cover **634** of the base portion **62** and extends inward in the pitch direction from the base portion **62** to the island-like portion **26** along the bottom portion **29** of the housing **20**. The protector **684**

extends from the middle portion **682** and covers the protected portion **28** of the island-like portion **26**.

[0078] Referring to FIGS. **9** and **10**, the protector **684** is located above the two movement-allowance portions **282** formed in the protected portion **28**. As previously described, the two movement-allowance portions **282** allow movements of the spring portions **668** of the two side arms **66** in accordance with resilient deformations thereof, respectively. The spring portions **668** have upper ends which are partially located in the movement-allowance portions **282**, respectively, even under an initial state where the spring portions **668** are not resiliently deformed.

[0079] As described above, the protector **684** of the present embodiment is arranged above the spring portions **668** so that the spring portions **668** are protected from above. The thus-arranged protector **684** prevents the spring portions **668** from being brought into abutment with the mating connector **70** during a process in which the mating connector **70** (see FIG. **6**) is mated with the connector **10**, for example. Thus, the present embodiment provides the connector **10** with a structure which prevents the spring portion **668** from buckling.

[0080] The protector **684** of the present embodiment is a metal plate and is hardly abraded even when it is brought into abutment with the mating connector **70** (see FIG. **6**). Accordingly, the spring portion **668** can be reliably prevented from buckling while abrasion of the resin of the housing **20** is prevented.

[0081] Referring to FIG. **13**, the protector **684** of the present embodiment is formed with two recessed portions **685**. Referring to FIG. **4**, each of the recessed portions **685** is located just over the spring portion **668**. In other words, the most part of the metal protector **684** is not located just over the spring portion **668**. However, referring to FIG. **4**, under a state where the spring portion **668** is not resiliently deformed, an upper end of the spring portion **668** is located inward in the width direction with respect to an outer edge **684E** of the protector **684** in the width direction.

[0082] According to the aforementioned arrangement, the mating connector **70** (see FIG. **6**) is brought into abutment with the protector **684** before being brought into abutment with the spring portions **668**. Therefore, the spring portions **668** can be effectively prevented from buckling. Moreover, since the protector **684** is formed with the recessed portions **685**, each of the spring portions **668** can be formed by easily bending a metal piece with no interference with the protector **684**. However, the present invention is not limited thereto. For example, the recessed portions **685** may be formed as necessary.

[0083] Referring to FIG. **11** together with FIG. **12**, the middle arm **68** of the present embodiment has an embedded portion **686** in addition to the middle portion **682** and the protector **684**. The embedded portion **686** is embedded in the island-like portion **26**. In detail, the embedded portion **686** is embedded in the protected portion **28**. The embedded portion **686** has a first portion **687** and a second portion **688**. The first portion **687** extends downward from the protector **684**. The second portion **688** extends outward in the pitch direction from the first portion **687**. In detail, the first portion **687** extends downward from an inner edge of the protector **684** in the pitch direction. The second portion **688** extends outward in the pitch direction from a lower end of the first portion **687**.

[0084] The protector **684** of the present embodiment is embedded in an upper end part of the protected portion **28** and is thereby held by the protected portion **28**. Referring to FIG. **2**, the protector **684** is located in the vicinity of the middle hole **27** of the island-like portion **26**. According to this arrangement, resin might be insufficiently poured around the protector **684** when the protector **684** is insert-molded in the housing **20**. As a result, the protector **684** might be insecurely held. In an instance in which the embedded portion **686** is not provided, the protector **684** might be detached from the protected portion **28** when a downward force is applied to the cover portion **63**.

[0085] In contrast, the embedded portion **686** of the present embodiment works as an anchor so that the protector **684** is securely held by the protected portion **28**. However, the present invention is not limited thereto. For example, the embedded portion **686** may be provided as necessary.

[0086] While there has been described what is believed to be the preferred embodiment of the invention, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such embodiments that fall within the true scope of the invention.

Claims

1. A connector comprising a housing and a hold-down, wherein: the housing has two long walls, two short walls, an island-like portion and a bottom portion; each of the long walls extends along a pitch direction; each of the short walls couples the long walls together in a width direction perpendicular to the pitch direction; the island-like portion is enclosed by the long walls and the short walls in a horizontal plane defined by the pitch direction and the width direction; the bottom portion is located at a lower side of the housing in an up-down direction perpendicular to both the pitch direction and the width direction; the bottom portion couples the island-like portion to each of the long walls and the short walls; the island-like portion has a protected portion; the protected portion is located at an end of the island-like portion in the pitch direction; the hold-down has a base portion, two side arms and a middle arm; the base portion has a cover portion; the cover portion covers the short wall from above; the two side arms are connected to opposite sides of the base portion in the width direction, respectively; each of the side arms has a side cover, a side portion, an extending portion, a spring portion and a contact point; for each of the side arms, the side cover covers the long wall from above, the side portion is connected to an outer part of the side cover in the width direction and extends along the pitch direction, the extending portion extends inward in the width direction from a lower end of the side portion, the spring portion extends upward from the extending portion so as to be resiliently deformable, and the contact point is supported by the spring portion; the middle arm has a middle portion and a protector; the middle portion extends inward in the pitch direction from the base portion; the protector extends from the middle portion and covers the protected portion of the island-like portion; the protected portion is formed with two movement-allowance portions; the two movement-allowance portions are recessed inward in the width direction and allow movements of the spring portions in accordance with resilient deformations thereof, respectively; the protector is located above the two movement-allowance portions; and each of the side arms is formed with a bent portion which is located at a boundary between a lower end of the side portion and the extending portion.
 2. The connector as recited in claim 1, wherein: each of the side arms has a coupling portion; and the coupling portion of each of the side arms couples the base portion and the side portion together.
 3. The connector as recited in claim 1, wherein: the middle arm has an embedded portion; the embedded portion is embedded in the island-like portion; the embedded portion has a first portion and a second portion; the first portion extends downward from the protector; and the second portion extends outward in the pitch direction from the first portion.
 4. The connector as recited in claim 2, wherein: the middle arm has an embedded portion; the embedded portion is embedded in the island-like portion; the embedded portion has a first portion and a second portion; the first portion extends downward from the protector; and the second portion extends outward in the pitch direction from the first portion.
-